General Anatomy and Physiology

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DMF Review Center

1. Which among these muscles move the mandible inferiorly?

Depression

✓A. External Pterygoid V_3

hx O: L.S of L.P.P of Sphenoid

✓B. Anterior belly of digastric V_3

C. Mylohyoid V_3

hx floor of the mouth muscle

D. Posterior belly of digastric VII

E. All of the above

2. Which cells will produce the hormone that will decrease serum calcium?

TAIL

Islets of Langerhans

A. Beta cells → Insulin

B. Alpha cells → Glucagon

GABI

C. Follicular cells of the thyroid → T3 T4

D. Parafollicular cells of the thyroid → Calcitonin

E. Chief cells of the parathyroid → PTH ↑ blood calcium

3. Which is NOT TRUE regarding Parathyroid Hormone (PTH)?

A. Its secretion will increase in response to decrease in serum calcium levels

B. It increases tubular secretion of calcium in kidneys

C. It increases absorption of calcium in the intestine

D. It increases serum calcium

E. None of the above

4. According to structure, the Temporomandibular joint is classified as?

"Ginglymoarthrodial"

A. Fibrous

B. Synovial

C. Cartilaginous

D. Synarthrosis

↳ immovable

E. Diarthrosis Freely movable

5. A depression at the surface of the bone

Oblique ridge :
DCR ML cusp
TR DB cusp

A. Process

↳ outgrowth or projection

B. Tuberosity

↳ rounded projection

C. Ridge

↳ linear elevation

D. Fossa

E. Notch

↳ border of bone

PPG

6. Which among these hormones is/are produced by the neurohypophysis?

X A. Prolactin by APG

↳ milk production

X B. Oxytocin by Hypothalamus

X C. Antidiuretic hormone by hypothalamus

X D. B and C only

E. None of the above

7. Which muscles are attached to the margins of the pterygomandibular raphe?

A. Masseter and Medial Pterygoid

→ form a sling around the mandible

B. Buccinator and Superior Pharyngeal Constrictor

C. Palatoglossus and Palatopharyngeus

→ pillars of the fauces

palatine tonsil

D. Buccinator and Masseter

E. Lateral pterygoid and Medial Pterygoid

→ L.P.P

8. Which nerve convey general sensory innervation to the anterior ⅔ (oral part) of the tongue?

A. Trigeminal

B. Facial

→ special A 2/3

C. Glossopharyngeal

→ General & Special P 1/3

D. Vagus

E. All of the above

9. Which endocrine organ is not affected by the signalling between the hypothalamus and hypophysis?

A. Thyroid

B. Adrenal cortex

C. Ovary

D. Pancreas

E. None of the above

1972

10. The following are vitamin K dependent clotting factors? EXCEPT

10, 9, 7, 2

✓ A. Stuart-Prower factor (10)

✓ B. Plasma thromboplastin component (9)

→ Hemophilia B

Vitamin K epoxide reductase = Warfarin

C. Antihemophilic factor (8)

→ Hemophilia A / 8

✓ D. Prothrombin (2)

E. None of the above

Intrinsic PTT

vs

Extrinsic PT

11. Which cells secrete the molecule essential for absorption of B12?

Intrinsic factor

Ileum

A. G cells

→ Gastrin

B. Chief cells

→ Pepsinogen → Pepsin

Pernicious anemia

Beefy tongue

C. Mucous neck cells

→ mucus

D. Parietal cells / OXYNTIC cells

→ IF + HCl

Strawberry tongue

Scarlet fever, Kawasaki dse

E. Enterocytes

→ fx unit of intestine

Chyme

12. Which hormone produced by the thyroid is more readily found in the circulation?

Propylthiouracil, methimazole

X thyroid peroxidase

X A. Thyroglobulin + iodine

↳ protein. precursor to thyroid hormone

X B. Thyrotropin releasing hormone TRH

↳ hypothalamus

T3 C. Triiodothyronine less abundant : more active"

↳ 20%

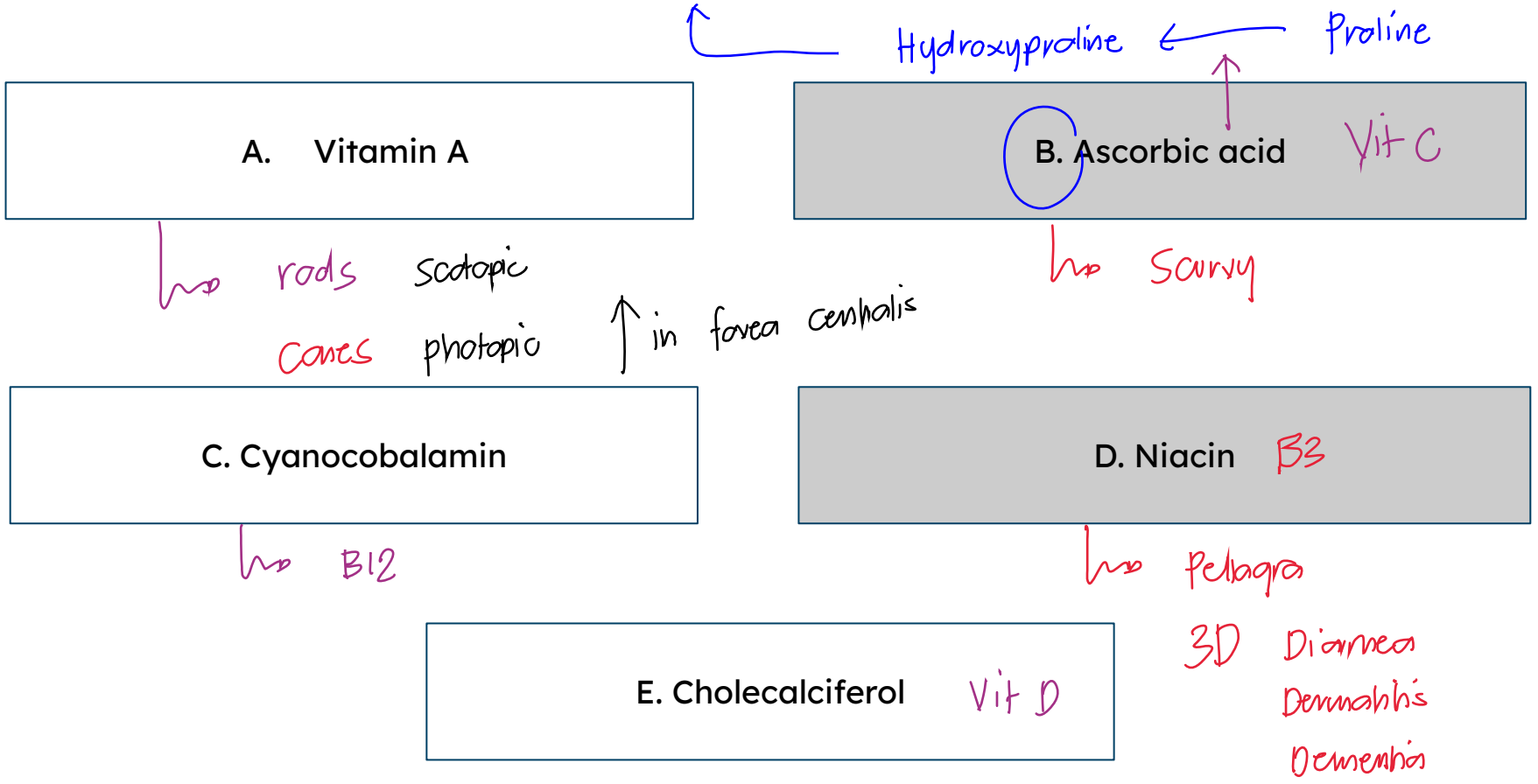
✓ T4 D. Thyroxine more abundant

↳ 80%

X E. Thyrotropin TSH

↳ APG

13. What vitamin is critical in collagen synthesis?



14. The amino acid that serves as precursor for catecholamines?

A. Tryptophan

→ Serotonin / 5-hydroxytryptamine

B. Tyrosine

→ epi, NE, dopa

C. Histidine

→ Histamine

D. Glutamic acid

① → ② Glutamate ⇝ ③ GABA

Excitatory

Inhibitory

E. Proline

→ collagen

15. Primary inhibitory neurotransmitter in the CNS?

A. Norepinephrine

B. Dopamine

C. Y-aminobutyric acid GABA

↳ ↑ influx of Cl⁻ ions

D. Serotonin

↳ mood

E. Acetylcholine

↳ Muscular contraction

16. Which of these vessels is not a branch of Internal Carotid artery?

→ terminal branch? ACA & MCA

✓ A. Anterior cerebral a.

✓ B. Middle cerebral a.

→ does not contribute to the Circle of Willis

C. Anterior communicating a.

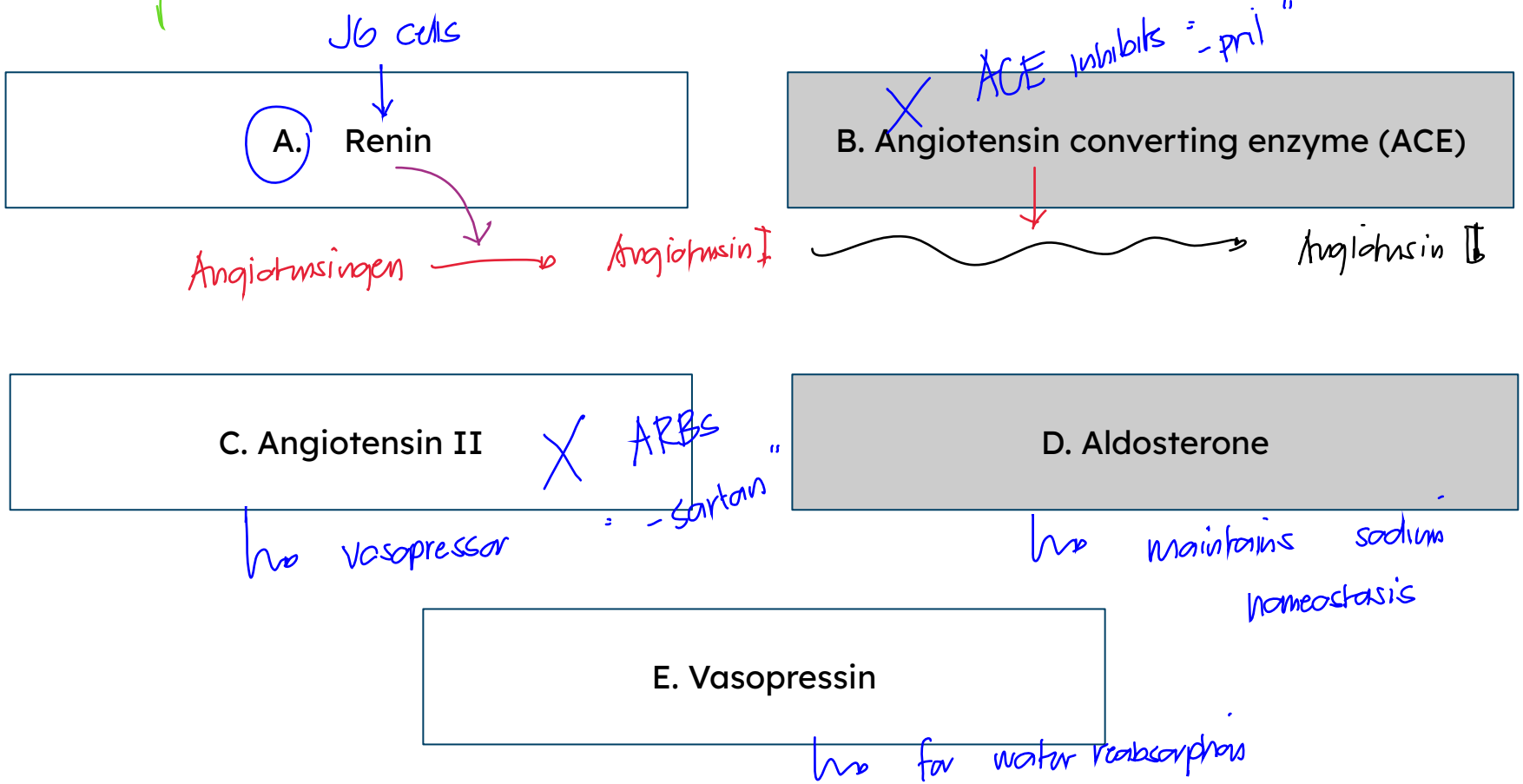
→ branch of : Anterior cerebral

✓ D. Posterior communicating a.

→ connects the ANT circ. to Post circulation

✗ E. C and D

17. In the RAAS pathway, angiotensinogen is converted to angiotensin I by the action of?



18. Which vasoactive amine is liberated by basophils?, mast cells, lungs

A. Epinephrine

lung VC

B. Norepinephrine

lung VC

VC = ↑ BP

C. Serotonin

D. Histamine

lung VD = ↓ BP

"shock"

Anaphylactic

DOC: epinephrine

E. None of the above

19. Cellular attachment found between the tooth surface and the junctional epithelium?

A. Tight junctions

↳ zonula occludens

B. Adherens junctions

↳ zonula adherens

autoimmune = pemphigus vulgaris

C. Desmosomes

Cells of same type

↳ macula adherens "spot weld"

autoimmune = Bullous pemphigoid

D. Hemidesmosomes

diff cell types

E. Gap junctions

↳ passage

"macula communicans"

"nexus"

20. Which statement among cardiovascular physiology is TRUE

A. Increased intracellular calcium improve myocardial contractility

↳ force of contraction

B. Release of epinephrine increases heart rate

↳ rate of contraction

C. Vagal stimulation decreases conduction velocity

D. Release of nitric oxide relaxes the smooth muscles of the blood vessels

↳ ↑ Vasodilation

E. All of the above

ISDN
Isosorbide dinitrate

21. Which chamber of the heart receives the blood returning from the pulmonary circulation?

Systemic C.

A. R. atrium

B. L. Atrium

C. R. Ventricle

D. L. Ventricle

E. None of the above

22. The following structure is NOT a feature found in the Right Atrium

main venous drainage of heart

A. Opening of the coronary sinus

B. Fossa ovalis

C. Opening of the pulmonary veins

in L.A

D. Opening of the superior vena cava

E. None of the above

23. This process involves movement of solvent, usually water, from low osmotic pressure to high osmotic pressure?

Solute ↑ = Osm. pressure ↑

A. Simple diffusion

↳ solutes from ↑ to ↓

B. Passive transport ✓

C. Facilitated diffusion

↳ carriers

OSMOSIS

D. ~~Posterior belly of digastric~~

ENDOCYTOSIS

E. ~~All of the above~~

↳ = "cell eating" PHAGO

or

= "cell drinking" PINO

24. The confluence of vertebral arteries at the base of the pons will form?

↳ transverse foramen C1 to C6

A. Common carotid a.

B. Basilar a.

Circle of Willis → A : ICA

Circle of Willis → P : V.A

C. Posterior cerebral a.

D. Superficial temporal a.

Aorta ~ E. Subclavian a. ~ Vertebral

25. Which is NOT TRUE regarding Insulin?

A. Its release is stimulated by decrease in blood sugar

B. It stimulates glycogenolysis

"glycogenesis"

Glycogen → Glucose

C. It stimulates gluconeogenesis

"inhibits"

Metformin

"liver"

D. It increases glucose levels in the blood

E. All of the above

26. Which is NOT TRUe regarding Cranial Nerve I OLFACTORY

A. It is purely sensory

B. Detects and process visual information

C. It exits the cranium through the ethmoid → cribriform plate

D. Damage may lead to anosmia

E. None of the above

= "phantosmia"

27. Which is NOT TRUe regarding Cranial Nerve II ~~no~~ OPTIC

A. It is purely sensory

B. It courses through the optic canal

sphenoid

~~no~~ ophthalmic a.

C. It received information from chemoreceptors X

photoreceptors

D. It does not influence the movement of the eyeballs

E. None of the above

28. Which is NOT TRUE regarding Cranial Nerve III ~ oculomotor

X A. It supplies the muscle that will move the eyeball laterally

→ L. rectus, CN VI

✓ B. Exits the cranium through the sphenoid

→ SOF

CN3, CNV1, CN6, CN4

✓ C. Exits with the smallest division of Trigeminal Nerve

"opthalmic"

✓ D. It is purely motor

E. None of the above

29. Which muscle is innervated by Cranial Nerve IV trochlear

A. Levator palpebrae superioris

→ CN III elevates eyelid

B. Orbicularis oculi

→ VII, closes eyelid

C. Lateral rectus

→ CN 6

D. Superior rectus

→ CN 3

E. None of the above S.O. 4

30. Which is NOT TRUE regarding Cranial Nerve V → trigeminal

A. The maxillary division is purely sensory ✓

B. The mandibular division supplies the muscle of mastication ✓

→ TIME

C. It is a mixed nerve ✓

D. Supplies the general sensation of pharyngeal part of the tongue ✗

Post 1/3 → IX

E. None of the above

31. Which is TRUE regarding Cranial Nerve VII FACIAL

A. Supplies the general sensation to the face

↳ CN V

B. Parasympathetic supply to the parotid gland

CN IX

↳ Sublingual, submandi

Lacrimal

C. Supplies the special sensation of pharyngeal part of the tongue

↳ IX

Should be "oral" part

D. It courses inside the substance of the parotid gland

E. All of the above

32. Which is NOT TRUE regarding Cranial Nerve VIII

~> Vestibulocochlear

A. It receives information from mechanoreceptors

Organ of Corti , Semicircular canals

B. It is purely sensory

C. Inputs are processed in the occipital lobe

visual input

X

temporal lobe ; Transverse Gyrus of Heschl

D. Provides information regarding balance

Cerebellum

E. None of the above

33. Which is NOT TRUE regarding Cranial Nerve IX → Glossopharyngeal

A. Transmits information regarding temperature and taste

B. Parasympathetic supply to the parotid gland

C. It is a mixed nerve

D. It is the sensory component of the gag reflex

→ motor: Vagus (X)

E. None of the above

34. Which is TRUE regarding Cranial Nerve X

Vagus

A. It supplies all palatal muscles

↳ t. veli palatini - CN V3

B. It supplies all pharyngeal muscles

↳ stylopharyngeus - CN IX

C. It supplies all muscles that will move the tongue

CN XII

↳ palatoglossus, only

D. It supplies all muscles that will change the shape of the tongue

↳ CN XII

↳ hypoglossal canal

E. None of the above

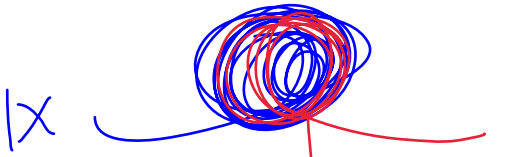
35. Which muscle will retract the tongue?

superiorly, posteriorly

A. Genioglossus

protrusion

B. Styloglossus



C. Hyoglossus

depression

D. Palatoglossus CN X

elevates tongue
depress palate

E. Geniohyoid

36. Which muscle is not innervated by a cranial nerve?

A. Trapezius

→ CN XI

B. Diaphragm ✓

→ Phrenic

C. Buccinator

→ VII

D. Mylohyoid

→ V3

E. Sternocleidomastoid

→ CN XI

37. The cells producing immunoglobulins are derived from?

plasma cells

= humoral "

A. Megakaryocytes

↓

Thrombocytes

B. B lymphocytes

↓

plasma cells

C. Monocytes

↓

macrophage

D. Mast cells

↳ releases: HISTAMINE

HEPARINS

E. T lymphocytes

= "cell mediated imm"

CD4 Helper

CD8 Cytotoxic

38. Cells that are constantly engaged in cell division is described as?

A. Stable cells

↳ upon injury

B. Labile cells

HYPERPLASIA

number

HYPERTROPHY

size

C. Permanent cells

↳ incapable once mature

D. Epithelial cells

E. Gametes

Cell division ? Meiosis

39. Which is NOT likely to happen in an increase in sympathetic tone?

"fight or flight"

A. Bronchodilation

B. Increased heart rate

C. Vasoconstriction

mydriasis

D. Miosis

pupillary constriction

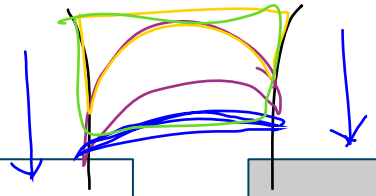
E. None of the above

40. How does the diaphragm behave during inspiration?

phrenic

A. Relaxes, moves inferiorly

B. Relaxes, moves superiorly



C. Flattens, moves inferiorly

D. Flattens, moves superiorly

Volume ↑ = Pressure ↓

E. Increases intrathoracic pressure

air moves ↑ pressure to ↓ press

41. Which is NOT TRUE regarding the trachea

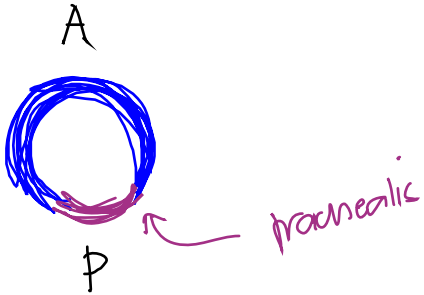
Conducting — nostril to terminal bronchiole
Respiratory — R. bronchiole alveoli

A. It is located anterior to the esophagus

B. Gas exchange begins at this site

C. It is supported by rings of hyaline cartilage

D. It bifurcates at the level of T4 → carina



E. All of the above

42. Which is NOT TRUE regarding the submandibular gland?

A. Frequently associated with salivary gland tumors

B. Frequently associated with sialolithiasis

Sublingual caruncle

parotid : Pleomorphic Adenoma

Mucoepidermoid carcinoma

C. Secretes a mixed predominantly serous secretion

D. It contributes to the greatest volume of saliva

E. None of the above

43. Amino acids being converted into glucose is a demonstration of what process?

A. Glycolysis

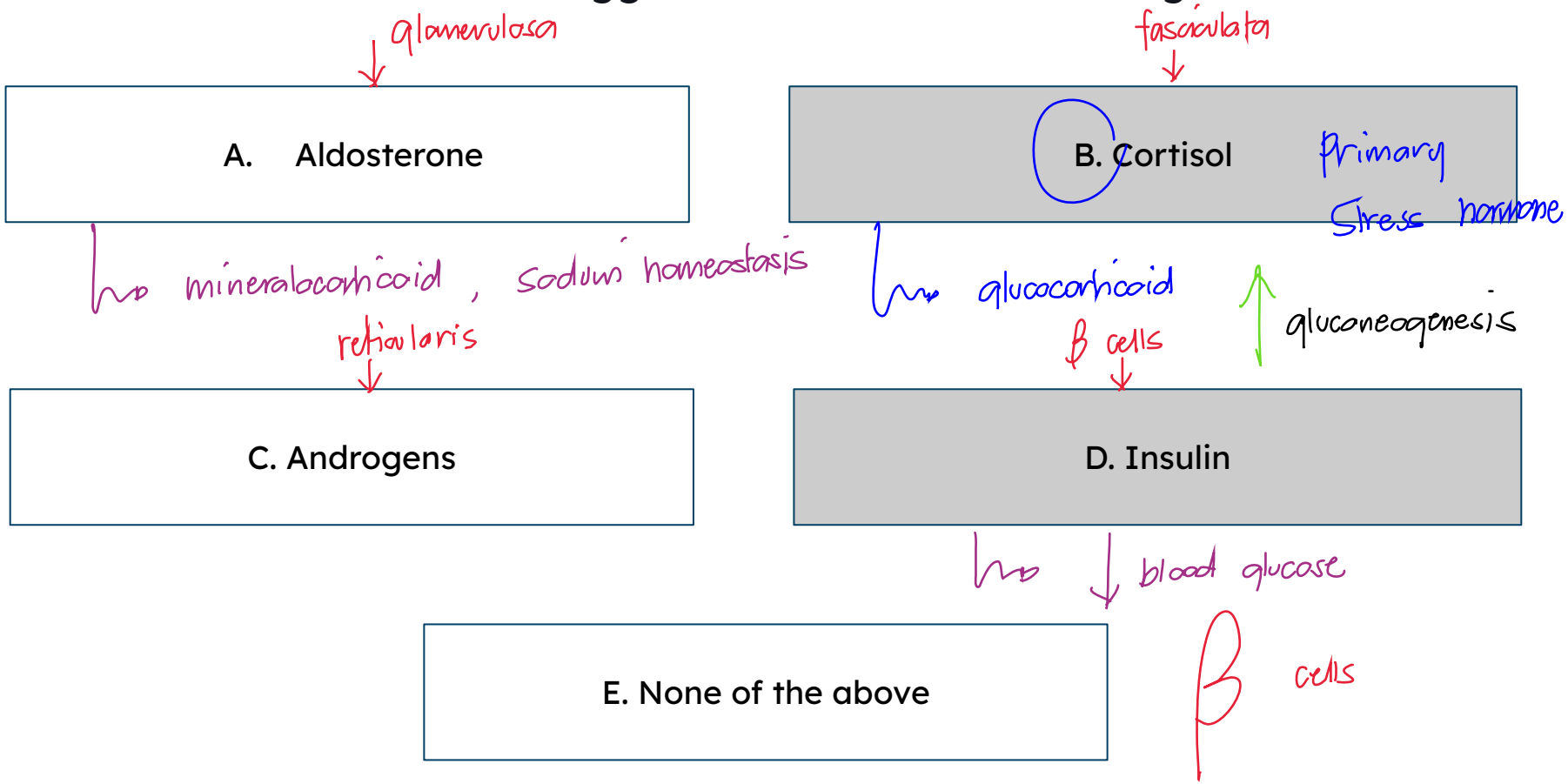
B. Glycogenesis

C. Gluconeogenesis

D. Glycogenolysis

E. Glucagon

44. The hormone that will trigger an increase in blood sugar?



45. Catecholamines are synthesized in what structure of the suprarenal gland?

A. Zona glomerulosa

↳ Aldosterone

B. Zona fasciculata

↳ cortisol

C. Zona reticularis

↳ androgens

D. Capsule

E. Medulla ↳ Chromaffin cells

46. Which is NOT TRUE regarding antidiuretic hormone / Vasopressin

A. It is antagonized by ethanol

B. It promotes vasoconstriction

in ↑ concentration

C. It promotes water reabsorption

↑ aquaporins

D. It is synthesized in the neurohypophysis

hypothalamus

E. None of the above

47. This medical term describes increased circulating thyroid hormones?

most common cause of hyperthyroidism

A. Graves' disease

↳ autoimmune stimulation of thyroid gland

B. Hypothyroidism

C. Hyperthyroidism

D. Thyrotoxicosis

E. Hashimoto's thyroiditis

↳ the most common cause of hypothyroidism in iodine suff. areas

48. What structure enables efficient transmission of action potential between cardiac myocytes?

Cardiac muscle tissue

A. Intercalated disks

allows cardiac myocytes to function as one unit

B. Gap junctions

↳ allows action potential to pass thru

Desmosomes Gap JX

C. Desmosomes

↳ mechanical linkage

D. Capillaries

↳ blood supply

E. SA node

↳ pacemaker of heart

49. TRUE about cell membrane

Plasmalemma

A. Also known as sarcolemma ✗

muscle

B. Also known as neurolemma ✗

neurons

C. Hydrophobic tail and hydrophilic head

↳ Phospholipid bilayer

✗ Semi-permeable

D. Impermeable to all substances

E. Provides structural framework to the cell ✗

↳ Cytoskeleton

50. The influx of sodium ions inside the neurons during depolarization is an example of?

A. Simple diffusion

B. Facilitated diffusion

C. Osmosis

D. Active transport

E. Endocytosis

51. Which event is expected in prolonged influx of chloride ions into the neuron?

GABA

A. Firing of action potential

B. A shift from negative to positive membrane potential

depolarization

Chloral hydrate = "knock out drops"

Benzodiazepines

antidote? Flumazenil

most common sedative used in dentistry

MIDAZOLAM

C. No action potential is produced

Hyperpolarization

(+)

D. Voltage gated calcium channels open

E. All of the above

52. A lingual papillae devoid of gustatory receptors?

A. Circumvallate

B. Foliate

C. Fungiform

Hyperkrophic ? Hairy tongue

D. Filiform

↳ most abundant

E. Sulcus terminalis

Hairy leukoplakia

↳ EBV / HHV

53. The zygomatic arch is formed between which 2 bones

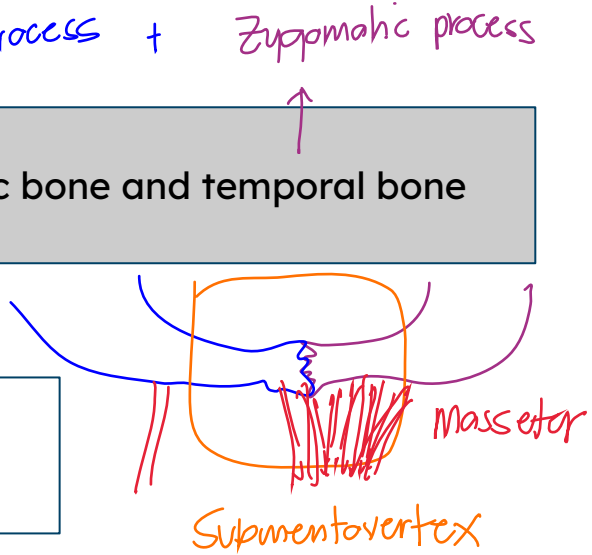
A. Zygomatic bone and maxilla

B. Maxilla and palatine bones

C. Zygomatic bone and frontal bone

D. Zygomatic bone and temporal bone

E. Temporal bone and sphenoid



54. The major venous drainage of the head and neck is via the?

A. Superior sagittal sinus

largest of dural venous sinus

B. Internal jugular vein

C. Coronary sinus HEART

D. Superior vena cava

R.A

E. Cavernous sinus

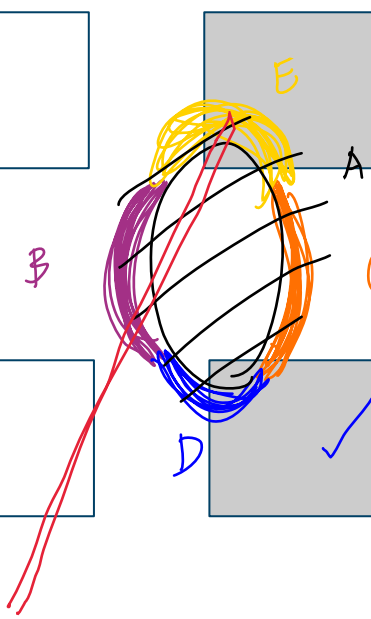
55. Which among these structures form the roof of the pterygomandibular space?

A. Lateral pterygoid

roof

B. Medial pterygoid

medial



diff. in max opening: TRISMUS

C. Ramus of the mandible

lateral

D. Pterygomandibular raphe

anterior

E. Parotid gland

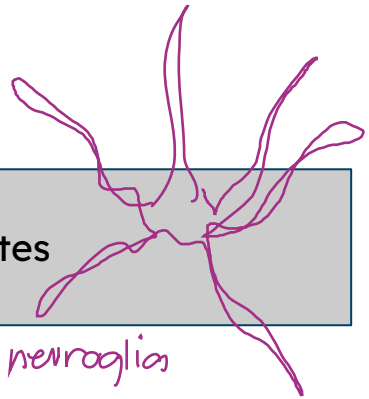
posterior

56. Resident macrophages of the nervous system tissue?

monocyte

A. Microglia

B. Astrocytes



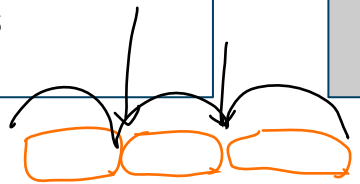
most abundant neuroglia

Blood brain barrier

C. Oligodendrocytes

CNS

myelin



D. Ependymal cells

CSF

blood CSF barrier

PNS

E. Schwann cells

57. Which is NOT TRUE about electrophysiology of neurons?

A. At rest, the neuron is permeable to sodium ions

B. The production of action potential is an ALL or NOTHING phenomenon

C. Blockage of sodium ion channels inhibits depolarization

D. Influx of sodium ions leads to depolarization

E. None of the above

58. Which is NOT TRUE regarding serotonin → mood

A. Available serotonin is decreased due to reuptake by serotonin transporters ✓

X

Sertraline

SSRI ; TCA

1st

B. Available serotonin is decreased due to breakdown by MAO ✓ X MAOi

↓

breakdown catecholamine

C. produced in the CNS ✓

→ raphe nuclei

D. produced in the digestive tract

→ gut ; enterochromaffin cells

E. None of the above

59. The medical term used to describe increased respiratory rate?

12 - 20 bpm / cpm

A. Respiratory depression

B. Hyperventilation

↳ CO_2 is expelled

Respiratory alkalosis

C. Tachypnea

D. Hypocapnia

↳ CO_2 in body

E. Tachycardia ↳ HR

60. Which refers to agents that influence the myocardial excitability?

A. Inotropic

↳ Force of contraction

Cardiac glycoside (+) inotropic

B. Chronotropic

↳ rate of contraction

C. Dromotropic

↳ Conduction velocity

D. Bathmotropic

E. None of the above

61. Which segment of the nephron is impermeable to water?

A. A. Proximal convoluted tubule

B. Descending limb of Loop of Henle

C. Thin ascending limb (TAL) of Loop of Henle

D. Distal convoluted tubule

E. Collecting ducts

62. Terminal branch of Internal Carotid Artery?

A. Anterior cerebral a.

+

MCa.

B. Anterior communicating a.

C. Superficial temporal a.

smaller

D. Maxillary a.

bigger

terminal br of :

ECA

E. Sphenopalatine a.

terminal br. of Maxillary a.

63. Protein synthesis takes place at which organelle?

A. Peroxisomes

B. Lysosomes

C. Nucleolus

D. Rough endoplasmic reticulum

RIBOSOMES ✓

✓

↳ ribosome synthesis

E. Smooth endoplasmic reticulum

64. Which is not a derivative of arachidonic acid in the cyclooxygenase pathway?

A. Prostaglandin

B. Prostacyclin

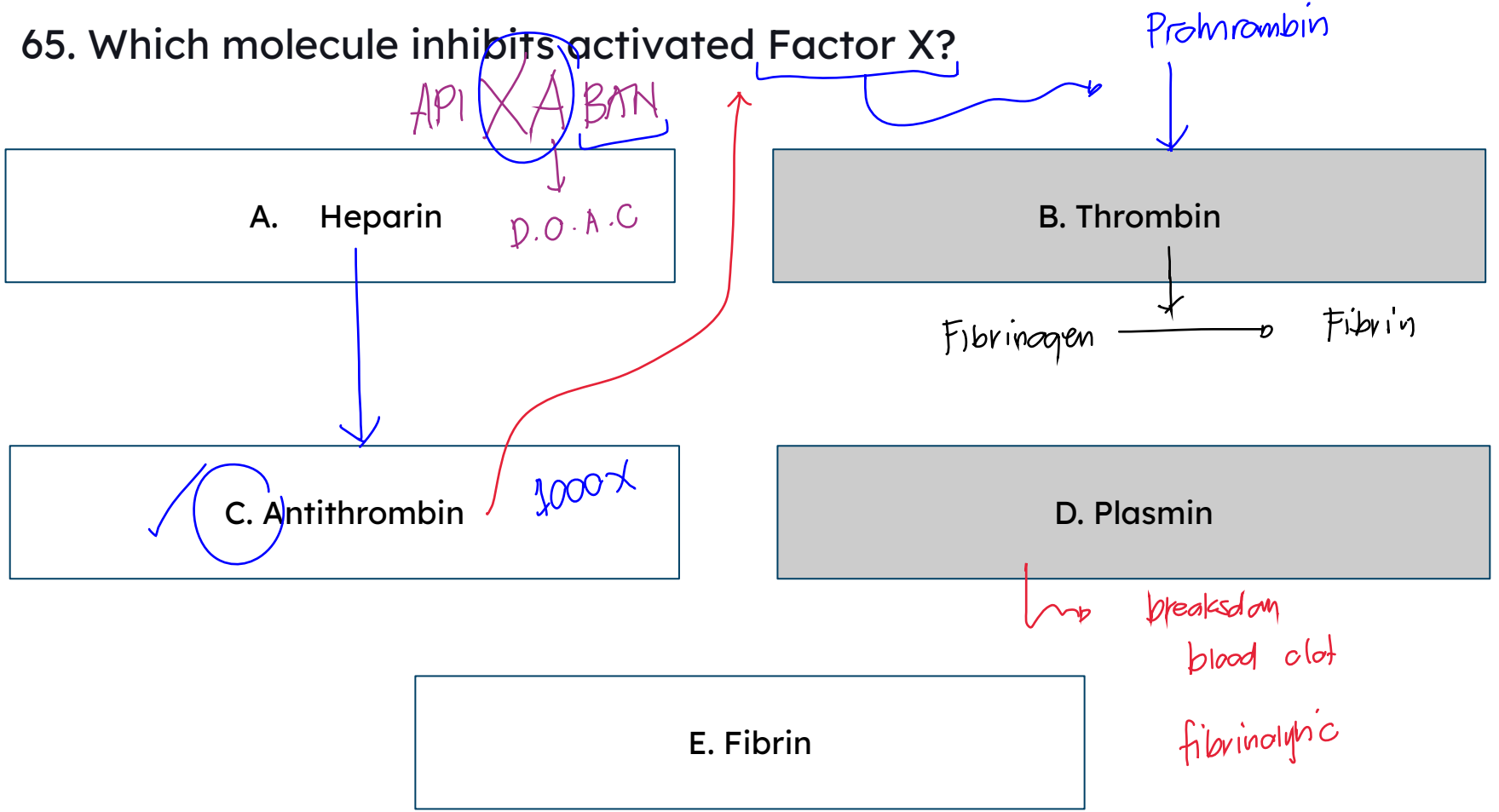
C. Thromboxane

D. Leukotriene

LIPOOXYGENASE Pathway

E. None of the above

65. Which molecule inhibits activated Factor X?



66. Which structure traverses the diaphragm together with the phrenic nerve?

② Esophageal hiatus T10

A. Esophagus + Vagus

③ Aortic hiatus T12

B. Aorta + TD

I 8 10 eggs At 12

C. Thoracic duct

D. Vagus

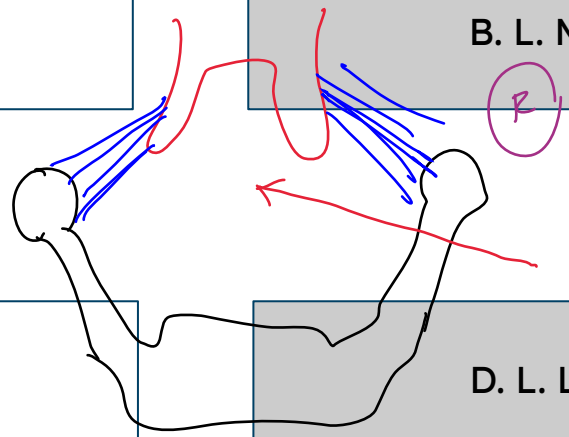
E. Inferior vena cava

① Vena caval hiatus - T8

67. The contraction of which muscle will move the mandible towards the left side?

A. R. Medial pterygoid

B. L. Medial Pterygoid



C. R. Lateral Pterygoid

D. L. Lateral Pterygoid

E. L. Temporalis

68. At what landmark will the Common carotid artery bifurcate into ICA and ECA?

A. At the level of C3-C5

Lup S.B of thyroid cartilage

B. At the superior border of cricoid cartilage

thyroid

C. At the level of T4

D. At the level of sternal angle

E. At the level of hyoid

69. The amount of blood pumped by the heart in one minute refers to?

mL / L

mL / min

/ min

A. Stroke volume

↳ amt of blood per beat / cont.

B. Heart rate

↳ cont / beat / min

C. Cardiac output

↳ CO

D. Systole

E. Diastole

$\frac{L}{beat} \quad \frac{beat}{min}$

70. The highest concentration of amino acid is found within?

A. Proximal convoluted tubule

B. Descending limb of Loop of Henle

A.A and glucose are absorbed in PCT 100%

C. Thin ascending limb (TAL) of Loop of Henle

PTH ↓ Ca^{2+}

D. Distal convoluted tubule ✓

No 25% of Na (X) Furosemide (Lasix)

E. Collecting ducts

71. The duct that emerges opposite the maxillary second molar belongs to?

A. Parotid gland

Stensen's

B. Von Ebner gland

purely serous

Granvalate

C. Submandibular gland

Wharton's

D. Sublingual gland

Bartholins, rivinus

E. Pancreas

Wirsung's

72. Responsible for milk ejection

letdown reflex

A. Prolactin

B. Oxytocin

↑ milk production

C. Vasopressin

D. Somatotropin

E. None of the above

73. Which among these statements regarding skeletal muscle contraction is/are NOT true?

A. Calcium must bind to nicotinic receptors

B. Acetylcholinesterase breaks down acetylcholine reducing its bioavailability

C. Calcium is stored and released from sarcoplasmic reticulum

D. Myosin is the thick filament while actin is the thin filament

E. None of the above

74. Refers to ion channels that respond to differences in membrane potential?

A. Leak channels

→ always open

B. Voltage gated channels

C. Ligand gated channels

→ needs a ligand / chemical

D. Mechanically gated channels

→ mechanical disturbances

E. Neurotransmitter

75. Which among these detects deep pressure?

A. Krause End bulbs

B. Merkel disks

C. Pacinian corpuscles

D. Free Nerve Endings

E. Carotid sinus